

3 (a) work done is force $\times$ distance moved in direction of force

or

no work done along PQ as no displacement/distance moved in direction of force B1

work done is same in vertical direction as same distance moved in direction of force B1 [2]

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- (b) (i) at maximum height $t = 1.5$ (s) or $s = \frac{1}{2}(u + v)t$, $s = 11$ m and $t = 1.5$ s C1
- $$V_v = 0 + 9.81 \times 1.5 \qquad V_v = (11 \times 2) / 1.5$$
- $$= 15 \text{ (14.7) ms}^{-1} \qquad \text{A1} \quad [2]$$
- (ii) straight line from (0,0) to (3.00, 25.5) B1 [1]
- (iii) at maximum height $V_h = 25.5/3 (= 8.5 \text{ ms}^{-1})$ B1
- $$\text{ratio} = mgh / \frac{1}{2}mv^2 \qquad \text{C1}$$
- $$= (2 \times 9.81 \times 11.0) / (8.5)^2$$
- $$= 3.0 \text{ (2.99)} \qquad \text{A1} \quad [3]$$
- (iv) deceleration is greater/resultant force (weight and friction force) is greater M1
- time is less A1 [2]